

GLADSTONE RADAR, Australia

WMO: 943800

Lat: 23.8553S

Lon: 151.2628E

Elev: 247

StdP: 14.57

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	52.7	54.9	27.3	21.4	62.0	31.7	26.3	61.8	22.4	69.3	20.2	68.4	8.6	220	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.3	91.1	76.9	89.4	76.0	87.9	75.1	79.9	87.2	78.7	85.1	77.6	83.9	11.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.0	147.0	83.6	76.9	141.6	82.6	75.8	136.4	81.5	43.7	87.6	42.5	85.2	41.3	83.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	21.5	19.5	48.2	95.8	2.0	2.8	46.7	97.8	45.6	99.4	44.4	101.0	43.0	103.0		
(5)			40.3	82.0	1.3	1.9	39.3	83.4	38.6	84.5	37.8	85.5	36.9	86.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.6	80.4	79.9	78.3	75.0	70.3	66.1	65.3	66.7	71.0	74.0	77.2	79.3
	DBStd	6.24	2.93	3.03	2.72	2.79	3.19	3.74	3.27	3.35	3.01	3.14	3.25	3.37
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	96	0	0	0	0	4	32	37	21	2	0	0	0
	CDD50	8608	942	837	878	749	628	483	473	519	630	745	815	909
	CDD65	3228	477	417	413	299	167	64	45	75	182	280	365	444
	CDH74	21034	4248	3596	2987	1449	444	85	62	157	648	1341	2382	3635
	CDH80	5763	1378	1127	790	245	23	1	1	6	71	253	681	1187
Wind	WSAvg	10.5	12.3	12.3	12.0	11.1	9.6	9.1	8.8	9.0	9.4	10.5	11.0	11.3
Precipitation	PrecAvg	36.6	5.8	6.1	4.0	1.9	2.2	1.5	1.4	1.3	0.9	2.2	2.8	5.2
	PrecMax	68.2	25.2	28.0	14.6	9.8	12.4	8.7	6.7	5.2	5.1	10.9	8.6	20.0
	PrecMin	17.0	0.2	0.3	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.1
	PrecStd	11.4	5.1	6.4	3.7	2.0	2.6	1.8	1.7	1.2	1.1	2.2	2.0	4.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.4	92.6	90.5	87.1	82.4	78.8	78.2	80.7	85.1	87.7	90.4	93.0
		MCWB	78.3	77.9	76.4	73.9	68.3	67.1	63.9	65.7	68.4	71.9	73.9	76.6
	2%	DB	90.8	90.4	88.1	84.4	80.0	76.4	75.7	77.5	81.9	84.7	87.9	90.3
		MCWB	77.4	77.4	75.3	72.2	67.6	65.4	63.4	64.2	67.5	71.6	73.6	76.0
	5%	DB	88.8	88.5	86.3	82.8	78.3	74.5	73.7	75.6	79.8	82.8	86.1	88.3
		MCWB	76.2	76.6	74.5	71.0	66.4	64.1	62.1	63.5	66.7	70.4	72.9	74.9
	10%	DB	86.9	86.5	84.5	81.1	76.6	72.7	71.7	73.7	77.9	81.0	84.2	86.3
		MCWB	75.3	75.4	73.4	69.8	65.6	63.1	61.0	62.2	65.9	69.3	71.7	74.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.6	81.4	79.7	77.7	73.7	70.6	69.4	70.2	73.5	76.1	78.7	80.9
		MCDB	89.4	89.1	86.2	82.8	78.0	74.9	73.2	76.1	80.1	83.8	86.7	89.3
	2%	WB	79.8	79.8	78.1	75.2	71.7	68.6	67.3	68.2	71.4	74.3	76.6	78.8
		MCDB	86.5	87.0	83.9	81.0	75.5	72.9	71.4	73.5	77.5	81.1	84.1	85.9
	5%	WB	78.4	78.3	76.7	73.3	69.8	66.9	65.7	66.6	69.9	72.8	75.1	77.4
		MCDB	84.3	84.8	82.8	79.3	74.9	71.5	69.6	71.4	75.8	79.1	82.1	84.1
	10%	WB	77.2	77.1	75.5	71.8	68.2	65.4	64.1	65.1	68.5	71.6	73.8	76.2
		MCDB	83.4	83.3	81.6	78.2	73.8	70.1	68.7	70.3	74.3	77.6	80.5	82.8
Mean Daily Temperature Range	5% DB	MDBR	13.3	13.0	13.0	13.3	13.6	13.4	14.5	15.4	15.7	14.8	14.4	13.9
		MCDBR	15.6	15.6	15.0	15.0	15.0	14.4	15.7	17.0	17.3	16.6	16.0	16.2
		MCWBR	5.9	6.2	6.2	6.3	7.2	7.3	8.0	7.9	7.6	6.9	6.3	6.4
	5% WB	MCDBR	13.6	13.8	13.0	13.4	12.6	12.3	13.3	14.4	15.1	14.9	14.8	14.5
		MCWBR	6.2	6.3	6.0	6.3	6.5	6.9	7.8	7.5	7.1	6.8	6.4	6.7
Clear-Sky Solar Irradiance	taub	0.389	0.386	0.372	0.344	0.322	0.315	0.305	0.316	0.353	0.374	0.391	0.391	
	taud	2.530	2.547	2.570	2.638	2.646	2.662	2.667	2.622	2.508	2.485	2.463	2.502	
	Ebn at Noon	303	299	295	291	285	281	288	295	296	300	300	303	
	Edh at Noon	36	34	32	28	25	24	24	28	34	36	38	37	
All-Sky Solar Radiation	RadAvg	2101	1949	1737	1525	1294	1120	1269	1541	1872	2067	2183	2184	
	RadStd	147	204	146	127	72	88	96	104	139	146	203	209	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page